

Disproof of Conjecture 000000001569

Arbitrarily large point sets with at most two angles

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Abstract

The conjecture asserts a uniform lower bound of $n^2 - O(n)$ for the number of distinct angles determined by an n -point set. Neither language of the original requires noncollinearity. We give an explicit n -point subset of the Euclidean plane with at most two angle values, for every n . These sets violate the claimed bound for every linear error constant and every asymptotic threshold. Lean 4 formalizes the actual Euclidean angles and the full quantified negation.

1 Statement and conventions

The English original states: “The angle set of a point set is the set of all angles it determines. Conjecture: An n -point set determines at least $n^2 - O(n)$ distinct angles; and the lower bound is attained by the equidistribution of arguments of lattice families.” The Chinese version has the same lower-bound assertion and supplies no additional restriction on the point sets. Both are included in `ORIGINAL.md`.

We use the ordinary planar Euclidean interpretation. For a finite set $S \subset \mathbb{R}^2$, let

$$\mathcal{A}(S) = \{\angle abc : a, b, c \in S \text{ are pairwise distinct}\}.$$

Here b is the vertex and $\angle abc \in [0, \pi]$ is the usual unsigned Euclidean angle between $a - b$ and $c - b$, in radians. Equal angle values are counted only once. All rays are nonzero. We include the collinear values 0 and π ; a convention excluding these values can only decrease our counterexamples’ angle counts. The source does not explicitly name the ambient dimension or orientation; our convention is stated rather than silently built into the argument. No general-position condition is assumed.

The standard uniform meaning of the lower-bound clause is

$$\exists C \in \mathbb{R}_{\geq 0} \exists N \in \mathbb{N} \forall n \geq N \forall S \subset \mathbb{R}^2 \text{ finite, } |S| = n \implies n^2 - Cn \leq |\mathcal{A}(S)|. \quad (1)$$

The constant C and threshold N are independent of both n and S . We disprove this clause. The additional lattice-attainment phrase is not precisely defined in the source and is not given an invented formalization: the failure of a necessary conjunct already refutes the stated conjunction.

2 The counterexample family

For each natural number n , set

$$S_n = \{(j, 0) : j \in \mathbb{Z}, 0 \leq j < n\}.$$

The first coordinate makes these n points distinct, so $|S_n| = n$. Every three points are collinear. At any vertex of a pairwise-distinct triple, the two nonzero rays either point in the same direction or in opposite directions. Their angle is respectively 0 or π . Therefore

$$\mathcal{A}(S_n) \subseteq \{0, \pi\}, \quad |\mathcal{A}(S_n)| \leq 2 \quad \text{for every } n \in \mathbb{N}. \quad (2)$$

This includes $n < 3$, where there are no qualifying triples.

Theorem 1. *For every real number C and every $N \in \mathbb{N}$, there exist $n \geq N$ and a finite planar set S with $|S| = n$ such that*

$$|\mathcal{A}(S)| < n^2 - Cn.$$

Consequently, (1) is false.

Proof. By the Archimedean property, choose $m \in \mathbb{N}$ with $m > \max(C, 2)$. Put $r = \max(N, m)$ and $n = r + 1$. Then $n \geq N$, $r > C$ and $r > 2$, hence $n - C > 1$ and $n > 3$. It follows that

$$n^2 - Cn = n(n - C) > n > 3 > 2 \geq |\mathcal{A}(S_n)|.$$

Taking $S = S_n$ proves the assertion. In particular it provides a strict violation for any nonnegative C and any threshold N proposed in (1), which is its negation. $\square$

3 Formalization and verification scope

The project pins Lean 4.19.0 and Mathlib v4.19.0. The manifest records every dependency revision. In `Solution.lean`, `Plane` is `EuclideanSpace` over the reals in dimension two. The finite set `angleSet` takes the image of all pairwise-distinct triples under Mathlib’s `EuclideanGeometry.angle`; the theorem `mem_angleSet_iff` proves its precise membership characterization. Thus the counted values are genuine angles, not a discrete surrogate.

The declarations `configuration_card` and `configuration_angleSet_card_le` establish the size and angle-count bounds. The geometry step proves the x-axis collinear and uses Mathlib’s collinearity characterization by angles 0 or π . The theorem `arbitrarily_large_counterexamples` proves the full statement above for every real C ; `not_uniformQuadraticLowerBound` negates (1). Finally, `original_conjunction_false` shows that adjoining any proposition for the attainment clause cannot make the conjunction true. It makes no independent claim about a particular lattice-equidistribution mechanism.

The submission supplies the full Lean source, pinned project, axiom inspection, reproducible verification instructions, and validation records. The proof is deductive and requires no numerical experiment or external computation. Local verification and independent semantic review are reported separately from any subsequent maintainer acceptance.

Source identity. The original is `conjectures/00000001569.md` at the repository commit below; the second identifier is its Git blob.

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9b795e7a94a6076e49a65a6489e1caf9153abd23
ec72f71c9bee352ef1bcb0201554f357dc7ff480
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